

Problem set 2

1. Let $f(x, y) = x^2 + y^2$ and let $D = [-1, 1] \times [0, 1]$.

(a) Evaluate $\iint_D f(x, y) \, dx \, dy$.

(b) Evaluate $\iint_D f(x, y) \, dy \, dx$ and compare it with the answer of (a).

$$(a) \iint_D f(x, y) \, dx \, dy$$

$$= \int_0^1 \int_{-1}^1 x^2 + y^2 \, dx \, dy = \int_0^1 \left[\frac{1}{3}x^3 + xy^2 \right]_{-1}^1 dy$$

$$= \int_0^1 \frac{2}{3} + 2y^2 \, dy = \left[\frac{2}{3}y + \frac{2}{3}y^3 \right]_0^1 = \frac{4}{3}$$

$$(b) \iint_D f(x, y) \, dy \, dx$$

$$= \int_{-1}^1 \int_0^1 x^2 + y^2 \, dy \, dx = \int_{-1}^1 \left[x^2y + \frac{1}{3}y^3 \right]_0^1 dx$$

$$= \int_{-1}^1 x^2 + \frac{1}{3} \, dx = \left[\frac{1}{3}x^3 + \frac{1}{3}x \right]_{-1}^1 = \frac{4}{3}$$

2. Let $g(x, y) = y(x^3 - 12x)$ and $D = [-2, 1] \times [0, 1]$.

(a) Evaluate $\iint_D g(x, y) \, dx \, dy$.

(b) Evaluate $\iint_D g(x, y) \, dy \, dx$ and compare it with the answer of (a).

$$(a) \iint_D g(x, y) \, dx \, dy$$

$$= \int_0^1 \int_{-2}^1 y(x^3 - 12x) \, dx \, dy = \int_0^1 y \left[\frac{1}{4}x^4 - 6x^2 \right]_{-2}^1 \, dy$$

$$= \int_0^1 \frac{57}{4} y \, dy = \left[\frac{57}{8} y^2 \right]_0^1 = \frac{57}{8}$$

$$(b) \iint_D g(x, y) \, dy \, dx$$

$$= \int_{-2}^1 \int_0^1 y(x^3 - 12x) \, dy \, dx = \int_{-2}^1 \left[\frac{1}{2} y^2 \right]_0^1 (x^3 - 12x) \, dx$$

$$= \int_{-2}^1 \frac{1}{2} (x^3 - 12x) \, dx = \frac{1}{2} \left[\frac{1}{4} x^4 - 6x^2 \right]_{-2}^1 = \frac{57}{8}$$

3. Compute the double integral $\iint_D \cos x \sin y \, dx dy$, where $D = [0, \pi/2] \times [0, \pi/2]$.

$$\iint_D \cos x \sin y \, dx dy$$

$$= \int_0^{\pi/2} \int_0^{\pi/2} \cos x \sin y \, dx dy$$

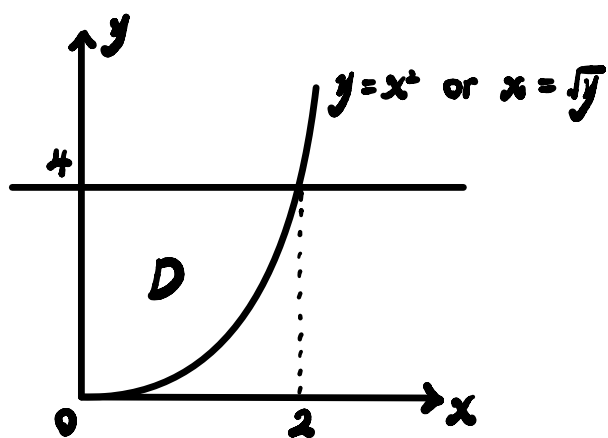
$$= \int_0^{\pi/2} [\sin x \sin y]_0^{\pi/2} dy$$

$$= \int_0^{\pi/2} \sin y \, dy$$

$$= [-\cos y]_0^{\pi/2}$$

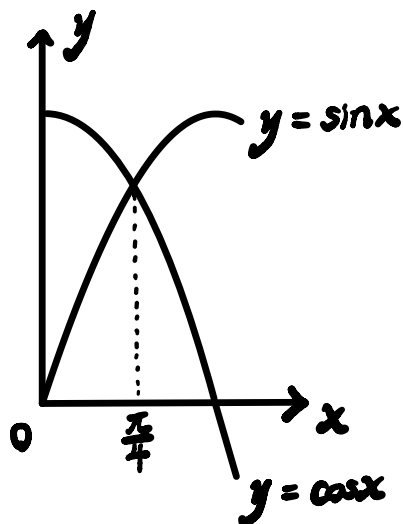
$$= 1$$

4. Evaluate $\iint_D x e^{y^2} dy dx$ if D is the region in the first quadrant bounded by the graphs of $y = x^2$, $x = 0$, $y = 4$.



$$\begin{aligned}
 \iint_D x e^{y^2} dy dx &= \iint_D x e^{y^2} dx dy \\
 &= \int_0^4 \int_0^{\sqrt{y}} x e^{y^2} dx dy \\
 &= \int_0^4 e^{y^2} \left[\frac{1}{2} x^2 \right]_0^{\sqrt{y}} dy \\
 &= \int_0^4 \frac{1}{2} y e^{y^2} dy \\
 \text{Let } u &= y^2, \quad \frac{du}{dy} = 2y \\
 &= \int_0^{16} \frac{1}{4} e^u du \\
 &= \left[\frac{1}{4} e^u \right]_0^{16} = \frac{1}{4} e^{16} - \frac{1}{4}
 \end{aligned}$$

5. A lamina has the shape of the region in the first quadrant that is bounded by the graphs of $y = \sin x$, $y = \cos x$, between $x = 0$ and $x = \pi/4$. Find its mass if the density is $\rho(x, y) = y$.



$$\begin{aligned}
 M &= \int \rho(x, y) dA = \iint_0 y dx dy = \int_0^{\pi/4} y dy dx \\
 &= \int_0^{\pi/4} \int_{\sin x}^{\cos x} y dy dx = \int_0^{\pi/4} \left[\frac{1}{2} y^2 \right]_{\sin x}^{\cos x} dx \\
 &= \int_0^{\pi/4} \frac{1}{2} [\cos^2 x - \sin^2 x] dx = \int_0^{\pi/4} \frac{1}{2} [1 - 2\sin^2 x] dx \\
 &= \frac{1}{2} \int_0^{\pi/4} 1 + \cos 2x dx = \frac{1}{2} \left[x + \frac{1}{2} \sin 2x \right]_0^{\pi/4} \\
 &= \frac{1}{2} \left(\frac{\pi}{4} + \frac{1}{2} \right) = \frac{\pi}{8} + \frac{1}{4}
 \end{aligned}$$

6. Let $f(x, y, z) = (x + 2y + 3z)^2$ and let $B = [0, 1] \times [-1/2, 0] \times [0, 1/3]$.

(a) Evaluate

$$\iiint_B f(x, y, z) \, dx \, dy \, dz$$

(b) Evaluate

$$\iiint_B f(x, y, z) \, dy \, dz \, dx,$$

and compare it with the answer of (a).

$$(a) \iiint_B f(x, y, z) \, dx \, dy \, dz$$

$$= \int_0^{\frac{1}{3}} \int_{-\frac{1}{2}}^0 \int_0^1 (x + 2y + 3z)^2 \, dx \, dy \, dz$$

$$= \int_0^{\frac{1}{3}} \int_{-\frac{1}{2}}^0 \left[\frac{1}{3} (x + 2y + 3z)^3 \right]_0^1 \, dy \, dz$$

$$= \int_0^{\frac{1}{3}} \int_{-\frac{1}{2}}^0 \frac{1}{3} (1 + 2y + 3z)^3 \, dy \, dz$$

$$= \int_0^{\frac{1}{3}} \frac{1}{3} \left[\frac{1}{8} (1 + 2y + 3z)^4 \right]_{-\frac{1}{2}}^0 \, dz = \int_0^{\frac{1}{3}} -\frac{1}{24} (3z)^4 \, dz$$

$$= -\frac{1}{24} \left[\frac{1}{15} (3z)^5 \right]_0^{\frac{1}{3}} = -\frac{1}{360}$$

$$(b) \iiint_B f(x, y, z) \, dy \, dz \, dx$$

$$= \int_0^1 \int_0^{\frac{1}{3}} \int_{-\frac{1}{2}}^0 (x + 2y + 3z)^2 \, dy \, dz \, dx = \int_0^1 \int_0^{\frac{1}{3}} \left[\frac{1}{6} (x + 2y + 3z)^3 \right]_{-\frac{1}{2}}^0 \, dz \, dx$$

$$= \int_0^1 \int_0^{\frac{1}{3}} -\frac{1}{6} (x - 1 + 3z)^3 \, dz \, dx = \int_0^1 -\frac{1}{6} \left[\frac{1}{12} (x - 1 + 3z)^4 \right]_0^{\frac{1}{3}} \, dx$$

$$= \int_0^1 -\frac{1}{72} x^4 \, dx = \left[-\frac{1}{360} x^5 \right]_0^1 = -\frac{1}{360}$$

7. Integrate e^{x+y+z} over the box $[0, 1] \times [0, 1] \times [0, 1]$.

$$\int_0^1 \int_0^1 \int_0^1 e^{x+y+z} dV$$

$$= \int_0^1 \int_0^1 \int_0^1 e^{x+y+z} dx dy dz$$

$$= \int_0^1 \int_0^1 \int_0^1 e^x e^y e^z dx dy dz$$

$$= \int_0^1 \int_0^1 (e^x e^y e^z - e^y e^z) dy dz$$

$$= \int_0^1 [(e^x e^z - e^z) - (e^z - e^z)] dz$$

$$= \int_0^1 (e^x e^z - 2e^z + e^z) dz$$

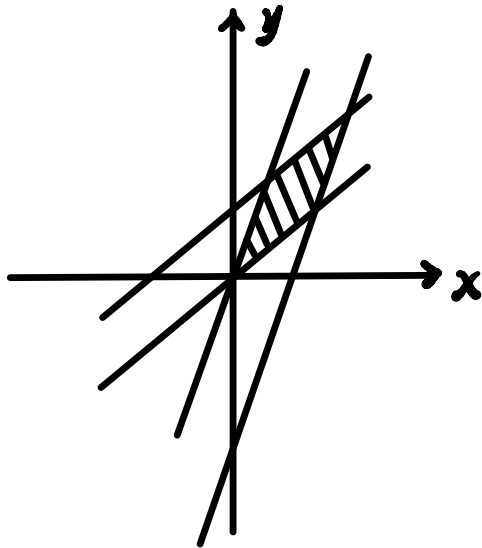
$$= (e^x e - 2e + e) - (e^x - 2e + 1)$$

$$= e^3 - 3e^2 + 3e - 1$$

8. Let P be the parallelogram bounded by $y = 2x$, $y = 2x - 2$, $y = x$, and $y = x + 1$. Evaluate $\iint_P xy \, dx \, dy$ by making change of variables

$$x = u - v, \quad y = 2u - v.$$

$$u = y - x, \quad v = y - 2x$$



$$y = 2x \rightarrow v = 0$$

$$y = 2x - 2 \rightarrow v = -2$$

$$y = x \rightarrow u = 0$$

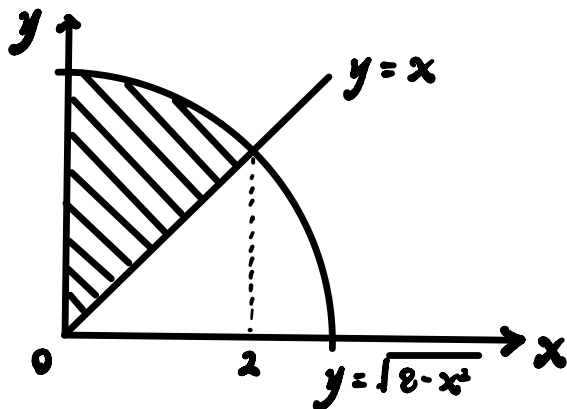
$$y = x + 1 \rightarrow u = 1$$

$$\begin{aligned} \therefore \iint_P xy \, dx \, dy &= \int_{-2}^0 \int_0^1 xy \left| \det \frac{\partial(x,y)}{\partial(u,v)} \right| \, du \, dv \\ &= \int_{-2}^0 \int_0^1 (u-v)(2u-v) \left| \det \begin{vmatrix} 1 & -1 \\ 2 & -1 \end{vmatrix} \right| \, du \, dv \\ &= \int_{-2}^0 \int_0^1 (2u^2 - 3uv + v^2) |-1+2| \, du \, dv \\ &= \int_{-2}^0 \left[\frac{2}{3}u^3 - \frac{3}{2}u^2v + uv^2 \right]_0^1 \, dv \\ &= \int_{-2}^0 \left(\frac{2}{3} - \frac{3}{2}v + v^2 \right) \, dv \\ &= \left[\frac{2}{3}v - \frac{3}{4}v^2 + \frac{1}{3}v^3 \right]_{-2}^0 = -\left(-\frac{4}{3} - 3 - \frac{8}{3}\right) = 7 \end{aligned}$$

9. Use polar coordinates to evaluate

$$\int_0^2 \int_x^{\sqrt{8-x^2}} \frac{1}{5+x^2+y^2} dy dx.$$

$$x = r \cos \theta, \quad y = r \sin \theta \quad \therefore x^2 + y^2 = r^2 \cos^2 \theta + r^2 \sin^2 \theta = r^2$$



$$\therefore \int_0^2 \int_x^{\sqrt{8-x^2}} \frac{1}{5+x^2+y^2} dy dx$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \int_0^{2\sqrt{2}} \frac{1}{5+r^2} r dr d\theta$$

$$\text{Let } u = r^2 \quad \therefore \frac{du}{dr} = 2r$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \int_0^8 \frac{1}{10+2u} du d\theta = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \left[\frac{1}{2} \ln |10+2u| \right]_0^8 d\theta$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{2} \ln \frac{13}{5} d\theta = \frac{1}{2} \ln \frac{13}{5} [\theta]_{\frac{\pi}{4}}^{\frac{\pi}{2}} = \frac{\pi}{8} \ln \frac{13}{5}$$

10. Find $\iiint_D e^{(x^2+y^2+z^2)^{3/2}} dV$ where D is the unit ball in $\mathbb{R}^3$.

for unit ball, $r=1 \quad \therefore x^2+y^2+z^2=1$

$$\iiint_0 e^{(x^2+y^2+z^2)^{3/2}} dV$$

$$= \int_0^\pi \int_0^{2\pi} \int_0^1 e^{r^3} \cdot r^2 \sin\phi \, dr \, d\theta \, d\phi$$

$$\text{Let } u = r^3 \quad \therefore \frac{du}{dr} = 3r^2$$

$$= \int_0^\pi \int_0^{2\pi} \int_0^1 \frac{1}{3} e^u \sin\phi \, du \, d\theta \, d\phi$$

$$= \int_0^\pi \int_0^{2\pi} \left(\frac{1}{3}e - \frac{1}{3} \right) \sin\phi \, d\theta \, d\phi$$

$$= \int_0^\pi \frac{2}{3}\pi (e-1) \sin\phi \, d\phi$$

$$= \frac{2}{3}\pi (e-1) [-\cos\phi]_0^\pi$$

$$= \frac{4}{3}\pi (e-1) (1+1)$$

$$= \frac{8}{3}\pi (e-1)$$